INTERNSHIP REPORT

Submitted

For the Partial Fulfilment of the Requirement for the Degree of Bachelor of _____ with _____ (Major and Minor Subjects) under **Magadh University**

Submitted to

The Department of _____, _____ **(College Name)**

[Paste College Logo here]

Submitted by

Name : _____

Semester : _____

Programme FYUGP : _____

University's Registration No. : _____

College Roll No. : _____

Year : _____

RK NEXORA
CAMPUS TO CAREER
A Unit of RKNexora Private Limited.

INTERNSHIP REPORT

Topic - Operations Management

Submitted to the Head of Department, _____ (College Name), in partial fulfilment of the requirements for the Degree of _____, Magadh University

Name of Student : _____

Programme : _____

University Registration No. : _____

College Roll No. : _____

Internship Registration No. : _____

Name of the IPO / Agency : RKNEXORA PRIVATE LIMITED

Internship Period : _____ to _____

Semester : _____

Year / Session : _____



_____ _____

Signature of Student **Seal & Signature of IPO**

ACKNOWLEDGEMENT

I express my sincere gratitude to RK NEXORA, RKNEXORA PRIVATE LIMITED, _____ (College Name), my faculty mentors and the internship supervisor, _____, for their guidance and support. The structured activities, learning resources and regular feedback helped me complete this internship in Operations Management successfully.

I am thankful to the Head of the Department and the college mentor for their encouragement and for guiding me in preparing this report. I am also grateful to the management of RKNEXORA PRIVATE LIMITED for providing a professional learning environment and the opportunity to learn and apply my knowledge in a practical setting. The concept sessions, hands-on tasks with sample data and case studies, and the supervisor's patient and detailed feedback helped me understand how organisations plan, run and improve the processes that produce goods and services.

Finally, I thank my family and peers for their constant encouragement throughout the programme.

_____

Name: _____

University Registration No. - _____

INTERNSHIP COMPLETION CERTIFICATE

The Internship Completion Certificate issued by RKNEXORA PRIVATE LIMITED for the internship in Operations Management is to be attached on this page.

[Attach the original / printed copy of the QR-verified Internship Completion Certificate here]

Certificate No. _____

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PREFACE

Internships are an important part of the undergraduate programme, as they allow students to see how classroom learning is applied in real organisations. The Four-Year Undergraduate Programme (FYUGP), introduced in line with the National Education Policy 2020, gives special importance to experiential learning, skill development and employability. This report has been prepared as part of the internship requirement of Magadh University and the Higher Education guidelines applicable in Bihar.

The report describes the internship undertaken in Operations Management at RKNEXORA PRIVATE LIMITED, the plan followed, the work carried out, and the learning gained. Behind every product and every service there is an operation - the set of activities that turns inputs into outputs. A tailor stitching a shirt, a hospital admitting a patient, a bank clearing a cheque and a factory assembling a motorcycle are all running operations, and their success depends on how well the work is planned, organised and improved. Operations management is the study of how to do this with less time, less cost, fewer errors and more satisfaction for the customer. It is useful in every kind of organisation, in manufacturing and services, large and small, and it is the foundation of careers in production, quality, logistics and general management. This report describes how the main areas of operations management were studied and how a garment-making unit was analysed for improvement using dummy data.

The report has been arranged in the order suggested by the University, beginning with the cover page and acknowledgement and ending with the conclusion and attendance sheet. It is hoped that it gives a clear and honest account of the internship experience. All examples in this report use dummy data and fictitious names; no real personal or business information has been handled.

EXECUTIVE SUMMARY

This report documents the internship completed in Operations Management at RKNEXORA PRIVATE LIMITED (RK NEXORA - 'Campus to Career'). The programme required a minimum of 120 hours, completed over about four to five weeks (28 working days) with daily sessions of four to five hours, and was supported by an attendance log, supervisor guidance and review.

Background. Operations management is the management of the process that converts inputs - labour, materials, machines, information and money - into outputs, which are goods or services, in the most efficient and effective way. Its main decisions concern the design of the process, the capacity, the quality, the inventory, the schedule, the cost and continuous improvement. Efficiency means using fewer resources for the same output, while effectiveness means producing what the customer actually needs. A well-run operation gives the customer a quality product on time at a fair price, and gives the organisation a profit.

Objectives. The internship aimed to (i) understand the role of operations and the input-process-output model, (ii) map processes and find bottlenecks, (iii) plan capacity, (iv) learn quality management and defect tracking, (v) learn the basics of lean thinking, 5S and waste, (vi) plan inventory, (vii) schedule work and manage workflow, (viii) measure productivity, (ix) control cost, and (x) find process gaps and suggest improvements.

Work carried out. Each topic had a concept session followed by hands-on exercises and calculations in MS Excel using dummy data and a fictitious garment-making unit. The student drew process flowcharts and found the bottleneck, calculated capacity utilisation, prepared check sheets and a Pareto chart of defects, analysed waste in a process, planned reorder quantities, prepared a work schedule, measured productivity, prepared a cost sheet and break-even, and proposed improvements. The later part of the programme was devoted to the project, a case study exercise, a mock practical exercise, supervisor review and preparation of this report.

Project. The work was tied together in a project titled **Process Improvement & Operations Study**. The study examined a fictitious garment-making unit, Shree Tailors and Garments, in Gaya: its process steps, bottleneck, capacity use, defects, waste, cost and productivity were analysed and a set of improvements with their expected results was proposed.

Method. A blended method was followed - learn, practise, document and review - supported by a daily attendance log with day-wise remarks and regular feedback from the supervisor.

Learning gained. The student gained a clear understanding of how operations are designed and run, practical skill in process mapping, capacity and productivity calculations, quality tools such as the check sheet and Pareto chart, lean waste identification and 5S, scheduling, cost sheet and break-even, and a structured method of improving a process.

Key observations. The output of a whole process is limited by its slowest step; a few causes usually account for most of the defects; waste is found by looking at the process from the customer's point of view; and improvements last only when they are standardised and measured.

Conclusion. The internship met its objectives and contributed to the student's academic understanding, employability and confidence. The report follows the University's suggestive format: cover page, acknowledgement, certificate, contents, preface, executive summary, about the agency, plan and execution, learning outcome, conclusion and attendance sheet.

ABOUT THE AGENCY

RK NEXORA, a unit of RKNEXORA PRIVATE LIMITED, provides structured internship and skill-development programmes under the theme 'Campus to Career'. The organization focuses on academic-industry alignment, practical exposure, assessment, documentation and employability-oriented learning.

Particulars	Details
Name of Agency	RKNEXORA PRIVATE LIMITED
Programme / Unit	RK NEXORA - Campus to Career
Registered Office	C/O Mukesh Kumar, Khiri, PS Atri, Jethiyan, Atri, Gaya - 805236, Bihar, India
CIN	U85499BR2026PTC086757
Email	rknexorapvtltd@gmail.com
Website	www.rknexora.org
Internship Topic	Operations Management

1. Vision

To support students in developing practical, professional and industry-relevant capabilities.

2. Mission

To deliver structured learning experiences that connect academic knowledge with workplace expectations.

3. Activities of the Agency

1. Structured internship programmes aligned with university and higher-education requirements.
2. Skill-development learning modules, practical activities and assignments.
3. Attendance tracking, supervision, assessment and documentation.
4. Employability-oriented guidance connecting campus learning to career readiness.

4. Programme Structure

The internship programme is organised around a fixed number of learning hours, completed over a defined period. Each intern is registered on the internship portal with a unique registration number, is assigned a supervisor, follows a day-wise plan and maintains an attendance log. Learning is organised into three phases: **Orientation** (Days 1-2), **Learning and Practice** (Days 3-22, ten sub-topics of two days each: one concept day and one hands-on day) and **Project and Review** (Days 23-28), followed by a concluding review.

5. Role of the Supervisor

The internship supervisor assigned by the agency guided the student through the programme, explained concepts, assigned practice tasks, reviewed the daily log and gave feedback. The supervisor also verified attendance and reviewed the student's work at the end of the programme.

PLAN AND EXECUTION

1. Objectives

The internship was planned around the following objectives, each linked to a part of the programme:

1. Understand the role of operations, and inputs, processes and outputs (Week 1).
2. Map processes with flowcharts and identify steps and bottlenecks (Week 1).
3. Plan capacity and measure utilisation and efficiency (Week 2).
4. Learn quality management, defect tracking, TQM and the Pareto principle (Week 2).
5. Learn lean basics - waste, 5S and continuous improvement (Week 3).
6. Plan inventory, scheduling and workflow, and measure productivity (Week 3).
7. Control cost, find process gaps and suggest improvements, and complete the project (Weeks 4-5).

2. Methodology

The programme followed a blended method made up of four connected steps, repeated for each topic:

Learn. Each topic began with a concept session in which the supervisor explained the idea, its main terms and how it is used in real organisations. Notes were prepared in the student's own words.

Practise. The concepts were then applied in guided hands-on tasks - drawing process flowcharts, calculating the capacity of each step and the bottleneck, calculating capacity utilisation and productivity, preparing check sheets and Pareto charts, listing the eight wastes in a process, calculating reorder quantities, preparing a work schedule, preparing a cost sheet and break-even, and writing an improvement plan, all in MS Excel. Only sample data and fictitious examples were used, so that no real personal or business information was handled.

Document. Every day's work was recorded in the internship attendance log with a short remark, and sheets, files and screenshots were saved so that the work could be reviewed.

Review. The supervisor reviewed the work, pointed out errors and suggested improvements. Review days, a case study and a mock practical exercise helped to consolidate the learning.

3. Understanding Operations Management - Concepts Studied

3.1 What are operations?

Every organisation exists to provide something of value to customers, and the operations function is the part that actually creates and delivers it. The operations of a factory are production; those of a bank are processing of accounts and transactions; those of a hospital are the care of patients; and those of a restaurant are cooking and serving food. In each case there are inputs, a transformation process and outputs, and a feedback loop from the customer that tells the organisation whether it is doing the right thing. Goods are tangible, can be stored and are made before they are sold, while services are intangible, cannot be stored and are produced and consumed at the same time, so that quality of service depends heavily on people and on the customer's participation.

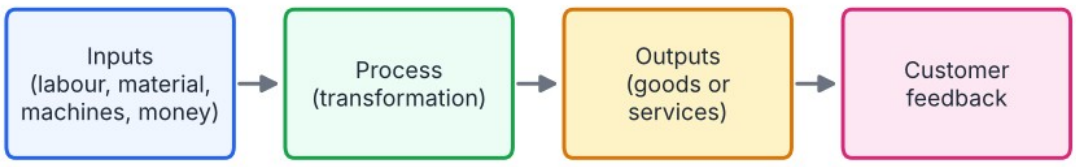


Figure 1: Inputs, process, outputs and feedback

3.2 The decisions of operations management

Operations managers take decisions at three levels. Strategic decisions are long-term - what to make, where to locate the plant, what technology and capacity to have. Tactical decisions are medium-term - the layout, the workforce, the quality system and the inventory policy. Operational decisions are daily or weekly - the schedule, the allocation of workers, the control of quality and the handling of breakdowns. The student studied the ten decision areas commonly listed in operations management: design of goods and services, quality, process and capacity, location, layout, human resources and job design, supply chain, inventory, scheduling and maintenance. In this internship the focus was on process, capacity, quality, inventory, scheduling and cost.

3.3 Process flow mapping and bottlenecks

A process is a sequence of steps that turns an input into an output. A process flowchart shows these steps with standard symbols - an oval for start and end, a rectangle for an activity, a diamond for a decision and arrows for the flow. Drawing the process makes it visible and shows delays, repeated work and unnecessary steps. The bottleneck is the step with the lowest capacity, because it decides how much the whole process can produce, however fast the other steps are. For a garment unit of the dummy firm, the time at each step for one garment is 2 minutes for order entry, 6 for cutting, 9 for stitching, 5 for finishing and 3 for packing.

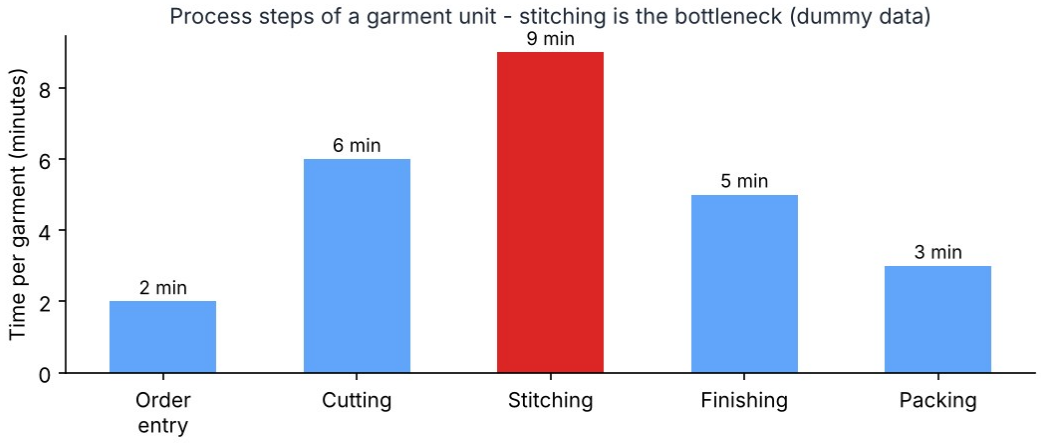


Figure 2: Time of each process step - stitching is the bottleneck

With one person at each step, the capacity of a step per hour is 60 divided by its time: cutting 10, stitching 6.7, finishing 12 and packing 20 garments. The stitching step, at 6.7 garments an hour, is the bottleneck and the whole unit cannot make more than that, which is about 53 garments in an eight-hour day. If a second stitching station is added, stitching capacity becomes 13.3 an hour, and the bottleneck moves to cutting at 10 an hour, which is 80 garments a day. The exercise showed an important rule: when a bottleneck is removed, another step becomes the new bottleneck, so improvement is a continuing activity.

3.4 Capacity planning

Capacity is the maximum output that a process can produce in a given time. Design capacity is the maximum under ideal conditions; effective capacity is the realistic maximum after allowing for breaks, maintenance, changeovers and so on; and actual output is what is really produced. Capacity utilisation is the actual output divided by the design capacity, and efficiency is the actual output divided by the effective capacity. For a dummy unit with a design capacity of 500 garments a day, an effective capacity of 450 and an actual output of 420, the utilisation is 420 / 500 = 84% and the efficiency is 420 / 450 = 93.3%. Capacity planning matches capacity to demand: too little capacity loses sales and delays deliveries, while too much capacity wastes money on idle machines and workers. Options for adjusting capacity include overtime, a second shift, subcontracting, adding machines and buying in finished goods.

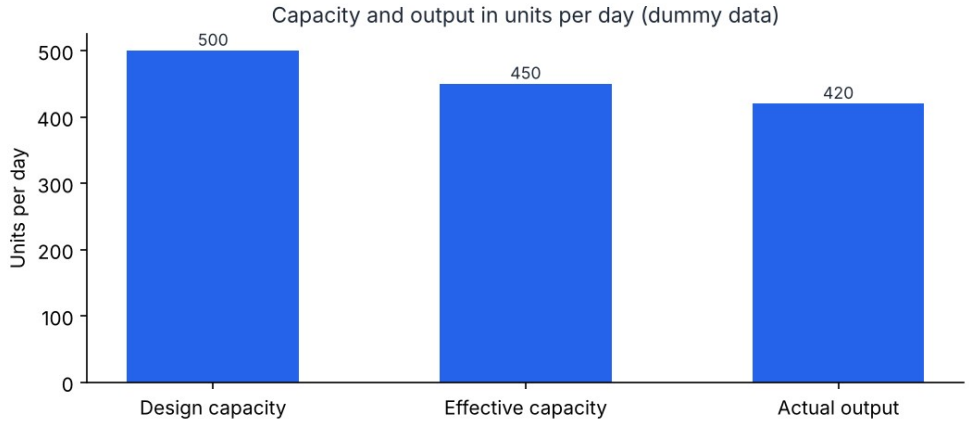


Figure 3: Design capacity, effective capacity and actual output

3.5 Quality management

Quality means that a product or service meets the customer's needs and the standards set for it. Quality is controlled by preventing defects, and not only by finding them at the end. Total Quality Management (TQM) makes quality the responsibility of everyone, focuses on the customer, uses facts and data and aims at continuous improvement. Inspection can be done on every piece (100% inspection) or on a sample, and the results are recorded on a check sheet. The seven basic quality tools were studied: the check sheet, the histogram, the Pareto chart, the cause-and-effect (fishbone) diagram, the scatter diagram, the control chart and the flowchart. In the Pareto chart, defects are ranked from the most to the least frequent, and the cumulative line shows that a few causes produce most of the problems.

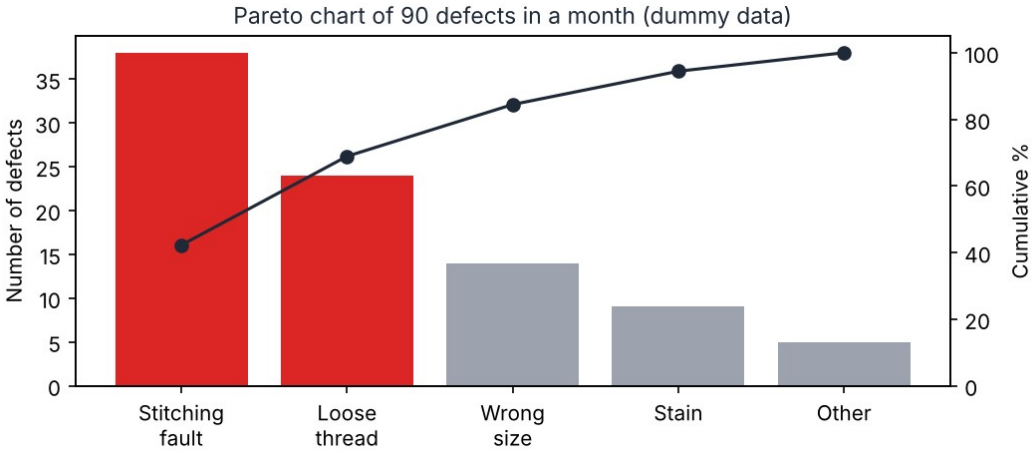


Figure 4: Pareto chart of defects in the garment unit

Of 90 defects recorded in a month, stitching faults (38) and loose threads (24) together account for 62 defects, which is 68.9% of the total, and adding wrong size (14) brings the share to 84.4%. The unit should therefore

concentrate on the stitching training and machine adjustment, and on thread trimming at finishing, where most of the improvement will come from.

3.6 Lean thinking, waste and 5S

Lean thinking, which began in the Toyota production system, aims to create more value for the customer with less waste. Waste is any activity that uses resources but adds nothing the customer would pay for. Eight wastes are commonly listed: defects, overproduction, waiting, non-utilised talent, transport, inventory, motion and extra processing. In the garment unit, waiting appeared when the stitching operators waited for cut pieces, motion when workers walked to a distant thread store, inventory when half-finished garments piled up before finishing, and defects in the form of re-stitching.

5S step	Meaning	Example in the unit
Sort	Remove what is not needed	Remove broken needles and old cloth pieces from the table
Set in order	A place for everything	Mark a place for scissors, thread and measuring tape
Shine	Clean and inspect	Clean machines daily and check them for oil leaks
Standardise	Make the rules a habit	Display a checklist and photographs of the correct arrangement
Sustain	Keep the discipline	Weekly 5S audit with scores

Continuous improvement (kaizen) means that small improvements, suggested by the people doing the work, are made every day. The cycle Plan, Do, Check, Act (PDCA) gives a method: plan the change, try it on a small scale, check the result and then adopt it if it works.

3.7 Inventory planning

Inventory in operations is of three kinds: raw materials, work in process (partly finished goods) and finished goods. Holding too much costs money and space, and holding too little stops production. The tools of inventory planning are the reorder point, safety stock and the economic order quantity (EOQ). For cloth, with a yearly need of 6,000 metres, an ordering cost of Rs. 400 per order and a holding cost of Rs. 10 per metre per year, EOQ = square root of (2 x 6,000 x 400 / 10) = square root of 480,000 = about 693 metres. This means about 8.7 orders a year, or one order every six weeks. With a daily use of 20 metres and a lead time of 5 days and a safety stock of 30 metres, the reorder point is (20 x 5) + 30 = 130 metres. The student noted that the EOQ is a guide, and that the order may be rounded to a convenient quantity, such as the roll length of the supplier.

3.8 Scheduling and workflow

Scheduling decides when each job is done and by whom. A good schedule meets the delivery dates, uses workers and machines well and keeps the work in process low. Simple loading rules were studied: first come first served, earliest due date first and shortest processing time first. For a set of jobs, the earliest due date rule reduces lateness, while the shortest processing time rule reduces the average waiting time. A Gantt chart shows the tasks as bars on a time scale, so that overlaps, gaps and delays are seen clearly. Workflow is the movement of work from one person or step to the next; it is improved by clear hand-over rules, visible queues (for example, a Kanban board with the columns To do, In progress and Done), limits on the work in progress and regular short meetings to remove blockages.

Job	Processing time (days)	Due date (day)	Order by earliest due date	Start	Finish	Late by
Job C	2	3	1st	0	2	0
Job A	3	6	2nd	2	5	0
Job D	1	7	3rd	5	6	0

Job	Processing time (days)	Due date (day)	Order by earliest due date	Start	Finish	Late by
Job B	4	8	4th	6	10	2

In this example, the earliest due date rule gives only one late job (Job B, two days late). The lesson was that a schedule must be checked against the promised dates, and that a late job should be discussed with the customer in advance.

3.9 Productivity measurement

Productivity is the ratio of output to input. Labour productivity is the output per labour hour, machine productivity is the output per machine hour, and multi-factor productivity compares the output with the combined cost of all inputs. For the dummy unit, an output of 480 garments with 80 labour hours gave a productivity of 6 garments per labour hour. After re-arranging the work and removing waiting time, the output rose to 540 garments with the same 80 hours, which gives 6.75 garments per labour hour, an improvement of 12.5%. Other measures are the cycle time, the lead time, the first-pass yield (the share of pieces correct the first time), on-time delivery and overall equipment effectiveness (OEE), which is the product of availability, performance and quality. If a machine has availability of 90%, performance of 95% and quality of 98%, its OEE is 0.90 x 0.95 x 0.98 = 83.8%.

3.10 Cost control

Cost control means knowing where the money goes and keeping it within the plan. Costs are classified as direct costs (material and labour that can be traced to a product) and indirect costs (overheads such as rent and electricity), and as fixed costs (which do not change with the quantity) and variable costs (which do). The cost sheet adds the elements to find the cost per unit and the profit. For one shirt, the dummy unit has a material cost of Rs. 180, labour cost of Rs. 70 and a variable overhead included in these, with a selling price of Rs. 360. The variable cost is Rs. 250, so the contribution per shirt is Rs. 110. With fixed costs of Rs. 60,000 a month, the break-even point is 60,000 / 110 = about 546 shirts. At 600 shirts, the revenue is Rs. 2,16,000 and the total cost is Rs. 1,50,000 + Rs. 60,000 = Rs. 2,10,000, so the profit is Rs. 6,000, which shows how small the margin is and why waste, rework and idle time matter.

Quantity (shirts)	Revenue (Rs.)	Variable cost (Rs.)	Fixed cost (Rs.)	Profit or loss (Rs.)
400	1,44,000	1,00,000	60,000	-16,000
546	1,96,560	1,36,500	60,000	About 0
600	2,16,000	1,50,000	60,000	6,000
800	2,88,000	2,00,000	60,000	28,000

3.11 Process improvement

Process improvement is a structured effort to make a process better. The steps studied were: define the problem and the goal; measure the present performance; analyse the causes with tools such as the fishbone diagram and the five whys; improve by trying solutions; and control by making the new method standard and measuring it - a cycle known as DMAIC, used in Six Sigma. For each improvement, the expected gain, the cost and the effort are estimated so that the easiest and most valuable ideas are done first. The student noted that people are the source of the best ideas, and that involving the workers and thanking them for suggestions is as important as the tools.

4. Week-wise Plan and Execution

Week	Days	Activities
Week 1	Days 1-6	Orientation and foundations: company induction, operations overview, process flow mapping and capacity planning, each with a study session and a practice task.
Week 2	Days 7-12	Quality and lean: quality management with check sheets and Pareto charts, lean basics, waste and 5S.
Week 3	Days 13-18	Planning and measurement: inventory planning, scheduling and workflow, productivity measurement.
Week 4	Days 19-24	Control and improvement: cost control, process improvement, the Process Improvement & Operations Study project, a case study exercise and a mock practical exercise.
Week 5	Days 25-28	Project closure: supervisor review and feedback, internship report drafting, presentation preparation and final review.

5. Week-wise Narrative

Week 1: The first week was devoted to orientation and foundations. After company induction, the student studied the input-process-output model, drew flowcharts of processes and found the bottleneck of a garment unit, and then calculated capacity, utilisation and efficiency.

Week 2: In the second week the student studied quality management and prepared a check sheet and a Pareto chart of defects. The week then moved to lean basics, in which the eight wastes were identified in a process and the 5S method was applied to a workplace.

Week 3: The third week was devoted to inventory planning with EOQ and reorder points, to scheduling with the earliest due date rule and a Gantt chart, and to productivity measurement.

Week 4: In the fourth week the student studied cost control and process improvement, completed the operations study for the garment unit, and carried out a case study and a mock practical exercise.

Week 5: The last week was devoted to project closure: supervisor review and feedback, drafting of the internship report, preparation of a presentation and final review before submission.

6. Topic-wise Learning in Detail

6.1 Operations Overview

Concept: The role of operations, inputs, processes and outputs. **Practice carried out:** Listing the inputs, process and outputs of five organisations - a school, a bank, a hospital, a bakery and a garment unit.

6.2 Process Flow Mapping

Concept: Drawing process flowcharts and identifying steps and bottlenecks. **Practice carried out:** Drawing the flowchart of garment making and finding the bottleneck from the step times (section 3.3).

6.3 Capacity Planning

Concept: Measuring capacity, demand and resource requirement. **Practice carried out:** Calculating utilisation and efficiency, and the workers needed for a demand of 100 garments a day.

6.4 Quality Management

Concept: Quality checks, standards, TQM and defect tracking. **Practice carried out:** Preparing a check sheet of defects, a Pareto chart and a fishbone diagram of 'stitching faults'.

6.5 Lean Basics

Concept: Types of waste, 5S and continuous improvement. **Practice carried out:** Marking the wastes on the process map and preparing a 5S checklist with scores for the workplace.

6.6 Inventory Planning

Concept: Stock requirement, EOQ and reorder planning. **Practice carried out:** Calculating EOQ, safety stock and reorder point for cloth and thread.

6.7 Scheduling and Workflow

Concept: Scheduling tasks, workflow allocation and tracking. **Practice carried out:** Scheduling four jobs by the earliest due date rule, drawing a Gantt chart and a Kanban board.

6.8 Productivity Measurement

Concept: Output, efficiency and productivity measures. **Practice carried out:** Calculating labour productivity before and after a change, and OEE of a machine.

6.9 Cost Control

Concept: Identifying cost heads and controlling expenses. **Practice carried out:** Preparing a cost sheet for a shirt and the break-even point (section 3.10).

6.10 Process Improvement

Concept: Finding process gaps and suggesting improvements. **Practice carried out:** Using the five whys on late deliveries and preparing an improvement plan with expected gains.

7. Good Practice, Safety and Ethics - Points Studied

Operations involve people, machines and money, so safety, fairness and honesty are part of good management. The points studied are summarised below.

- **Safety first:** machines must have guards, workers must be trained and the workplace kept tidy; the Factories Act, 1948 and the Occupational Safety, Health and Working Conditions Code lay down the duties.
- **Fair treatment of workers:** fair wages, reasonable working hours, rest breaks and no child or forced labour.
- **Quality honesty:** defects must be reported and corrected, not hidden; standards must not be lowered to meet a target.

- **Respect for customers:** promise only the delivery date and quality that can be given.
- **Environmental care:** reduce waste, use materials carefully, dispose of waste properly and save energy.
- **Data honesty:** measures such as output and defects must be recorded truthfully, because decisions depend on them.
- **Involvement of people:** workers' suggestions are valued and acknowledged.

8. Process Flows Studied

8.1 Improvement cycle (PDCA)

- Plan: describe the problem, find its causes and decide the change to try.
- Do: try the change on a small scale and record what happens.
- Check: compare the result with the plan and the earlier performance.
- Act: if it worked, make it standard and train everyone; if not, learn why and plan again.

8.2 Finding the cause of a problem with five whys

- Why were the shirts delivered late? - Because stitching was not finished.
- Why was stitching not finished? - Because the stitchers waited for cut pieces.
- Why did they wait? - Because cutting was done once a day in a big lot.
- Why in a big lot? - Because the cutting table is shared and set-up takes long.
- Action: cut in smaller lots twice a day and keep a buffer in front of stitching.

9. Worked Examples with Sample (Dummy) Data

Worked example 1 - effect of adding a stitching station. The step capacities of the garment unit (garments per hour) before and after adding a second stitching station were compared. The output of the unit is set by the lowest capacity.

Step	Time per garment (min)	Capacity before (per hour)	Capacity after (per hour)
Order entry	2	30	30
Cutting	6	10	10
Stitching	9	6.7	13.3
Finishing	5	12	12
Packing	3	20	20
Output of the unit		6.7	10 (cutting is the new bottleneck)

The daily output rises from about 53 garments (6.7 x 8) to 80 garments (10 x 8), a gain of 27 garments a day or about 50%. At a contribution of Rs. 110 per garment, the extra contribution is 27 x 110 = Rs. 2,970 a day. If the second station costs Rs. 25,000 (machine and a worker's wage for a month, say), it would be recovered quickly, but only if cutting is also run well, since cutting is now the limit. The next improvement should therefore be at cutting.

Worked example 2 - the effect of waste on cost. From the defect Pareto chart, 90 defects were recorded in a month of 600 shirts, which is a defect rate of 15%. If each defect costs Rs. 40 to repair (labour and thread), the monthly cost of defects is 90 x 40 = Rs. 3,600. Halving the stitching faults and the loose threads (62 defects) would save about Rs. 1,240 a month in repair cost, in addition to better delivery and fewer returns. This is small in money, but it is worth doing because it is cheap, needs only training and trimming at the end of stitching, and also improves the reputation of the unit.

Worked example 3 - OEE of the stitching machine. The machine was planned to work 8 hours (480 minutes) but stopped for 48 minutes for breakdowns and changeovers, so the availability is (480 - 48) / 480 = 90%. It produced 410 shirts against an ideal output of 432 in the running time, so the performance is 410 / 432 = 94.9%. Of these, 400 were good, so the quality is 400 / 410 = 97.6%. The OEE is 0.90 x 0.949 x 0.976 = 83.4%. The world-class target is generally 85%, so the unit is close, and the largest loss is availability, which suggests a planned maintenance routine.

10. Project: Process Improvement & Operations Study

Problem statement. Shree Tailors and Garments, a fictitious unit with 14 workers in Gaya, makes school uniforms and shirts. Orders are often late in the school season, defects have to be repaired, and the owner feels that the workers are busy but the output is low. The aim of the project was to study the operation and to suggest improvements.

10.1 Method

- Draw the process flow and measure the time at each step; find the bottleneck.
- Calculate the capacity, utilisation, efficiency and productivity.
- Record defects for a month and prepare a Pareto chart.
- Observe the unit for wastes and make a 5S assessment of the workplace.
- Prepare a cost sheet and the break-even point.
- List improvements with their expected gain and effort, and decide the order of implementation.

10.2 Findings

Area	Finding	Evidence
Bottleneck	Stitching at 9 minutes per garment limits output to about 53 garments a day	Process timing (Figure 2)
Capacity	Utilisation 84% and efficiency 93.3%	420 actual, 500 design and 450 effective
Quality	Stitching faults and loose threads cause 68.9% of the defects	Pareto chart (Figure 4)
Waste	Waiting for cut pieces, walking to the thread store, piles of half-finished garments	Observation
Productivity	6 garments per labour hour	480 garments in 80 hours
Cost	Break-even at about 546 shirts; profit only Rs. 6,000 at 600 shirts	Cost sheet

10.3 Improvement plan and expected results

Improvement	Effort	Expected result
Add a second stitching station (or a part-time operator) and cut in two lots a day	Medium	Output from 53 to about 80 garments a day
Train stitchers on the two most common faults and trim threads at finishing	Low	Defects reduced by about half; repair cost saved
Move the thread and tools next to the stitching tables (5S)	Low	Less walking and waiting; productivity up by about 10%
Weekly planned maintenance for machines	Low	Availability up from 90% towards 95%
Use a Kanban board for orders and a weekly Gantt schedule	Low	On-time delivery above 90% in the school season

10.4 Limitations

The study used a limited period of observation and dummy figures. A real study would need more observations of times, a check of the machine and worker data, the cost of the changes and the opinion of the workers. The student noted that the numbers are for learning, and that real improvements should be tried on a small scale first and measured before they are adopted.

11. Case Study Exercise: Long Waiting Time in a Small Diagnostic Centre

A fictitious diagnostic centre in Patna offers blood tests and X-rays. Patients complain that they wait for more than 90 minutes on busy mornings, although each test itself takes 10 to 15 minutes. Staff say they are overloaded, while the manager notes that the equipment is idle for long periods in the afternoon. The student was asked to analyse the operation as a service process.

Process mapping. The patient's path was drawn: registration, payment at the cash counter, waiting, sample collection or X-ray, waiting for the report and report collection. The time was measured at each step on a sample of 20 patients. Registration and payment took 8 minutes on average but the waiting before sample collection was 55 minutes, and the largest delay was at the cash counter where one person handled both registration and payment.

Analysis. The bottleneck is the front desk, not the laboratory. The arrivals are bunched between 8 and 10 in the morning because most patients are fasting for blood tests, so there is a queue at the first step and idle time later in the day. Many patients also arrive without the doctor's request or the documents.

Proposed solution. Separate registration and payment into two counters in the morning peak; offer booked time slots, with the first slots for fasting tests; send a message listing the documents to bring; start a token system with a display; and move some X-ray bookings to the afternoon. The waiting time and the number of patients per hour should be measured every week. The expected benefit is a waiting time of about 30 minutes, and the cautions noted were to keep patient data confidential and to avoid reducing the quality of checks in order to save time.

12. Mock Practical Exercise: Analysing and Improving a Small Process

A mock practical exercise was conducted at the end of the learning phase to check that the student could analyse and improve a process independently. The task was to study the process of a fictitious photocopy and printing shop that takes 40 minutes to complete an average order. The steps followed were:

1. Process map: the steps from receiving the order to delivery were drawn and the time of each step was noted.
2. Bottleneck: the slowest step (waiting for the single large-size printer) was identified.
3. Waste: waiting, motion and rework were marked on the map.
4. Quality: the errors of one week were tallied on a check sheet and a Pareto chart was drawn.
5. Cause analysis: the five whys were used for the most common error.
6. Improvement: a second printer slot for small jobs, a check before printing and a tidy work table (5S) were proposed.
7. Measures: expected order time, error rate and productivity were estimated.
8. Presentation: the analysis and the plan were explained to the supervisor in about five minutes.

13. Sample Checklists Prepared

Process analysis checklist

- Process drawn with all steps and times measured, not guessed.
- Bottleneck and its capacity identified.
- Wastes marked on the map.
- Data on defects, delays and rework collected for at least one week.

5S and quality checklist

- Unneeded items removed; a place and a label for every tool and material.
- Workplace cleaned and machines inspected daily; checklist displayed.
- Defects recorded on a check sheet and reviewed weekly.
- Improvements standardised, documented and audited every week.

14. Tools and Resources

The following tools and resources were used during the internship: MS Excel for capacity, productivity, cost, break-even, Pareto charts and Gantt charts, MS Word and MS PowerPoint for notes and presentation, templates for process maps, check sheets, 5S audits and cost sheets, case studies and learning material given by the agency, reference notes on safety and labour laws, and regular mentor feedback.

15. Programme at a Glance

The programme was in the domain of Operations Management with RKNEXORA PRIVATE LIMITED as the agency. It required a minimum of 120 hours over 28 learning days, organised in three phases: Orientation (Days 1-2), Learning and Practice (Days 3-22) and Project and Review (Days 23-28). Attendance was recorded in the Internship Attendance Log, signed by the supervisor (see Attendance Sheet).

The approximate distribution of the effort across the topics is shown below.

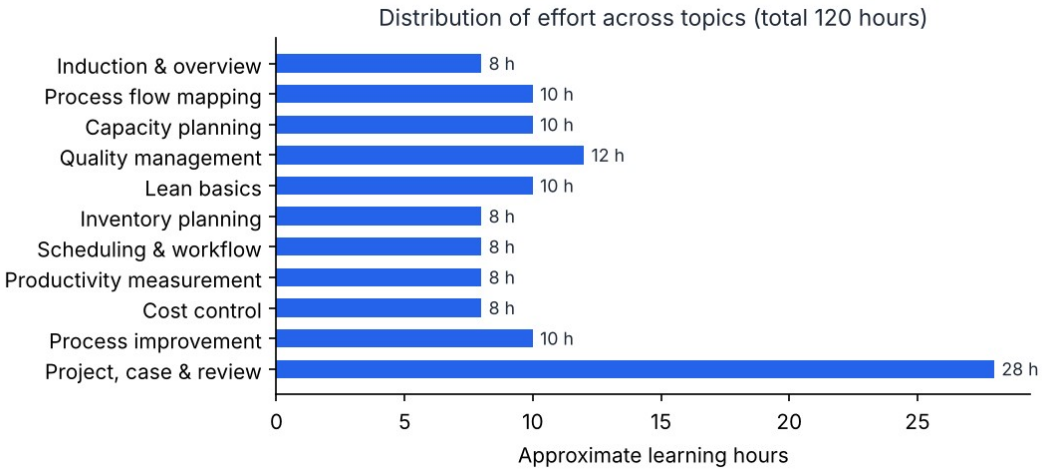


Figure 5: Distribution of learning effort across topics (120 hours)

LEARNING OUTCOME

a. Knowledge Acquired

1. The meaning of operations, the input-process-output model and the decisions of operations management.
2. Process flowcharts, cycle time, bottleneck and the effect of removing a bottleneck.
3. Design capacity, effective capacity, utilisation and efficiency.
4. Quality, TQM, the seven quality tools, check sheet and Pareto chart.
5. Lean thinking, the eight wastes, 5S, kaizen and PDCA.
6. EOQ, safety stock and reorder point for inventory planning.
7. Scheduling rules, Gantt chart, Kanban and workflow.
8. Productivity and OEE measurement.
9. Cost classification, cost sheet, contribution and break-even.
10. A structured method for process improvement using DMAIC and the five whys.

Glossary of Terms Learned

- **Operations** - The activities that convert inputs into goods or services.
- **Process** - A series of steps that produce an output.
- **Bottleneck** - The step with the lowest capacity, which limits output.
- **Cycle time** - The time to complete one unit at a step.
- **Capacity** - The maximum output in a period.
- **Utilisation / Efficiency** - Actual output divided by design / effective capacity.
- **TQM** - Total Quality Management.
- **Pareto principle** - A few causes produce most of the effects.
- **Check sheet** - A form for recording defects as they occur.
- **Lean / Waste (muda)** - Producing value with less waste / activity that adds no value.
- **5S** - Sort, Set in order, Shine, Standardise, Sustain.
- **Kaizen** - Continuous improvement.
- **PDCA / DMAIC** - Improvement cycles.
- **Kanban** - A visual board that shows the status of work.
- **Gantt chart** - A bar chart of tasks against time.
- **Productivity** - Output divided by input.
- **OEE** - Overall Equipment Effectiveness (availability x performance x quality).
- **Contribution** - Selling price minus variable cost per unit.
- **Break-even point** - The quantity at which revenue equals total cost.

b. Skills Learned

Technical skills. Drawing process flowcharts; calculating cycle time, capacity, utilisation, efficiency and productivity; preparing check sheets, Pareto charts and fishbone diagrams; identifying wastes and applying 5S; calculating EOQ and reorder points; preparing schedules, Gantt charts and Kanban boards; preparing a cost sheet and break-even; and using MS Excel for calculations and charts.

Professional skills. Observation and analytical thinking; structured problem solving; attention to detail and accuracy; practical judgement; communication with workers and managers; time management; and presenting improvements in simple words.

These skills were developed through repeated practice and daily documentation. Measuring real-looking processes step by step built observation and accuracy, using the quality tools built a habit of using facts, and explaining the findings to the supervisor built confidence and communication.

c. Observed Attitude and Gained Values

1. **Respect for people:** workers are the source of the best ideas, and improvements must be made with them and not against them.
2. **Honesty with data:** facts and measurements must be recorded truthfully.
3. **Commitment to quality and safety:** quality and safety must never be traded for speed.
4. **Continuous learning:** small improvements every day add up.
5. **Discipline and punctuality:** regular reporting at fixed timings and daily attendance records.

d. Challenges Faced

Finding the real bottleneck instead of the step that looked busiest was handled by measuring the time of each step and calculating capacities.

Understanding the many terms such as utilisation, efficiency and OEE was handled by working each formula with one numerical example.

Drawing the Pareto chart correctly was handled by sorting the defects and checking the cumulative percentage.

Separating the causes from the symptoms was handled by using the five whys and the fishbone diagram.

Making practical suggestions that are low in cost was handled by estimating the effort and expected gain of each idea.

Managing time between calculations, documentation and report writing was handled by dividing work week by week and using review days for corrections.

e. Career Pathways

The knowledge gained supports careers such as Operations Executive, Production Planning and Control Executive, Quality Assurance Executive, Process Improvement Assistant, Lean and 5S Coordinator, Inventory and Scheduling Executive, Branch or Store Operations Manager and Junior Business Analyst. Opportunities exist in manufacturing, food processing, hospitals, banks, retail, logistics, hotels, e-commerce and government services. Related higher-study options include MBA or PGDM in operations, BBA, and certificate courses in lean, Six Sigma, quality management and project management. A good next step is to learn Excel and a process-mapping tool and to take a Lean Six Sigma white or yellow belt certificate.

CONCLUSION

The internship in Operations Management was a valuable learning experience that integrated academic knowledge with practical activity. The programme achieved its objectives through structured learning, guided practice, assignments, supervisor review and documentation.

The student gained a clear understanding of how organisations plan, run and improve their operations - through process mapping, capacity planning, quality management, lean thinking, inventory planning, scheduling, productivity measurement and cost control - and acquired practical skill in the calculations and tools used for each, applied in a process improvement study. The experience has strengthened the student's employability and provides a foundation for further study and work in operations and quality management.

Reflection

Looking back, the most useful lesson of the internship was that a process is limited by its slowest step and that the best improvements come from looking at the real process with real measurements. Workers at the busiest step were not necessarily the problem; the step before them was. A few causes explained most of the defects, and small changes such as keeping tools near the work saved a surprising amount of time. Using simple tools such as flowcharts, Pareto charts and the five whys showed that improvement does not need complicated technology, but it does need careful observation and honest data.

Objectives and Achievement

The objective of understanding operations and mapping the process was achieved through the input-process-output figure and the process flowchart with the bottleneck. The objective of capacity and quality was achieved through the utilisation calculation and the Pareto chart. The objective of lean, inventory and scheduling was achieved through the waste and 5S assessment, the EOQ calculation and the schedule with the Gantt chart. The objective of productivity, cost and improvement was achieved through the productivity, OEE, cost sheet and break-even calculations and the improvement plan of the garment unit.

Future Scope

The knowledge gained can support further study of operations, quality and supply chain management, and careers in production, quality and process improvement. With the growth of manufacturing, food processing, logistics and service industries in Bihar and across India, and the focus on productivity and quality in the 'Make in India' initiative, people who can improve operations are increasingly needed.

Suggestions for Improvement of the Programme

1. Include a visit to a factory or service centre with appropriate permissions to observe a real process.
2. Provide practice with process-mapping and simulation tools.
3. Add a group project in which students improve a real process of a local small business.
4. Continue regular mentor feedback and progress reviews.
5. Encourage students to maintain detailed daily documentation.

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ATTENDANCE SHEET

The Internship Attendance Log issued by the IPO, with the supervisor's signature for each day, is to be attached on this page.

[Attach the signed Internship Attendance Log here]